PhysiqueQuest

Existing features and functionality

Implementation inventory | 29 October 2026

Scope and reading guide

This inventory describes code present in the workspace on 29 October 2026, including uncommitted work. It is a code inspection, not proof that every provider or deployment is operational.

The draw.io document uses linked-by-label off-page references A-G and C1. A node groups related operations; it is not a sequence diagram or a guarantee that all branches are reachable in a particular deployment.

1. Launch, web and mobile

Next.js supplies the user interface; the Expo app hosts it in a WebView and supplies native camera, network, device and notification bridges. Connection and permission states affect availability.

Navigation exposes Home, Coach, Scan, Quests and Me. Session restoration and profile/onboarding determine what can be shown.

2. Authentication and account sessions

Register and sign in through session endpoints. Email verification, verification codes, password reset and password change have request and confirmation paths.

Sessions can be refreshed or migrated, listed, revoked, logged out individually or logged out everywhere. Access-protected APIs use server-side authentication.

3. Profile, onboarding and settings

Profile data includes personal measurements, goals, training preferences, health information and an emergency contact. User-facing forms update profile data; photo upload and removal are supported.

Settings include workout and safety-related preferences. The backend stores profile and settings against the account; reads and saves can fail and must surface errors.

4. Home, goals and progress

The dashboard aggregates activity, nutrition and workout information, including missions and progress links. Goals and milestones summarize account-derived values.

Progress supports recording, listing and deleting measurements such as weight, waist and bench press. Values and strength estimates depend on user-provided records.

5. Coach and exercise library

The Coach plan is derived from the profile and available exercise catalog. Users can browse the exercise library, search, filter by muscle and equipment, page through results, and open instructions and demonstrations where available.

Recent catalog queries and guide details use a bounded account-scoped device cache. The server remains authoritative. Administrator catalog operations are protected separately.

6. Prepare and video analysis

Live camera readiness includes connection and coarse local scene checks. Camera permission and device support affect availability; a user can continue without the camera.

The upload flow previews MP4 or MOV locally and sends it only after Analyze video. The backend validates type, size and duration and forwards it to a configured pose provider. MediaPipe estimates landmarks and joint angles.

An optional selected squat or chest press can yield an experimental local movement-cycle count when signal checks pass. Gemini review, when configured and available, may supply a likely exercise, equipment, target muscles, estimated repetitions, observations, corrections and instructions.

Tracking coverage is not a form score. Correct-form repetitions, injury risk and overall accuracy are not validated. Provider failures can leave the review unavailable; the report includes retry and does not automatically modify the workout.

7. Workout session and report

Create or recover an active workout, record sets with repetitions and weight, pause/resume, and complete the session. Saved account data and device-only drafts have different recovery behavior.

Completion creates a workout report based on saved workout data. Sharing may be unavailable in Expo Go. Workout history and summaries feed related dashboard and activity views.

8. Nutrition, food scan and dietary plan

Add food through search or manual entry, or select/capture a food image and review an estimated scan before saving. Users can edit/delete meals, view today and history, and update water intake.

My Dietary Plan provides active targets, custom goals and meal allocation. Recommendations may use a configured AI provider and are reviewed before saving. Sample nutrition is explicitly separate from account data.

9. Quests, activity and leaderboard

Activity tracks account-derived missions and engagement state. Quest screens display missions, streaks, levels and badges based on available data.

Leaderboard endpoints supply quest and strength views, subject to eligibility and privacy rules. Strength rankings use self-reported sets and estimated values rather than independently verified lifts.

10. Notifications and reminders

The notification page lists system items, supports read/unread and mark-all-read, and stores reminder preferences. The mobile app includes device registration and notification permission handling.

Native notification presentation and delivery depend on device permission, provider configuration and deployment. A registered device or visible notification screen does not by itself prove production delivery.

11. AI assistant and safety

Phiq chat sends a user question and relevant account summaries to the configured Gemini integration and returns a text response when available.

Safety endpoints create and resolve alerts. Inactivity monitoring and emergency-related behavior depend on configuration, device state, permissions and provider readiness; do not assume dispatch is guaranteed.

12. Administration, privacy and backend services

Admin endpoints expose analytics and alert review and support exercise catalog management subject to authorization. The engagement screen includes a preview path; do not interpret preview data as live production analytics.

Account APIs support export, deletion and recovery flows. Supporting services include signed storage uploads/downloads, nutrition analysis jobs, synchronization snapshots, health and readiness endpoints. Their presence does not establish that every UI path uses them.

13. Data, cache and dependencies

MongoDB stores account-associated server records. The device cache holds selected bounded resources, is scoped to account/API, expires data, refreshes and invalidates entries; it does not cover every page or guarantee offline writes.

External dependencies include configurable pose analysis, Gemini, food search/nutrition providers, media storage and push infrastructure. Availability, cost, permissions, data retention and deployment settings affect behavior. Secrets stay in backend configuration.

Code evidence

Primary sources: backend/src/app.module.ts; backend/src/*/*.controller.ts; frontend/sections/Home, Coach, Scan, Quests Section and Me Section; frontend/lib/api.ts and deviceCache.mjs; mobile/App.tsx; pose-service/main.py.

Detailed flow evidence: frontend/sections/Coach/CoachSection.tsx and CameraCheck.tsx; frontend/sections/Scan/ScanSection.tsx; backend/src/pose/pose.service.ts and video-review.service.ts; backend/src/nutrition/nutrition.controller.ts; backend/src/notifications/notifications.controller.ts.